

REVIEW

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A Narrative Review of Effects and Underlying Mechanisms of Needling Therapy for Treating Myofascial Pain and Musculoskeletal Disorders

Cheuk Ming Tong^{1*}

Abstract

This review aims to summarise the effects and underlying mechanisms of needling therapy (NT) in the treatment of myofascial pain and musculoskeletal (MSK) disorders. This review aims doing so by categorising NT effects according to different anatomic levels of the human body and linking each effect to the specific mechanism. Acupuncture, the first form of NT, has existed for 3000 years and been practised by Traditional Chinese Medicine (TCA); however, its mechanism is hard to be validated by modern science as it involves “Chi”, “Yin & Yang” and “meridians”—all the ambiguous concepts which are hard to be interpreted by the West and modern science. Dry needling (DN), another form of NT, which emerged in the West in the twentieth century, has not well been explicitly reviewed for its mechanisms. Given that NT—acupuncture originated in the East and DN originated in the West—has been proven by systematic reviews and meta-analyses for its effectiveness to treat MSK disorders (world's major disability contributor as defined by World Health Organization) and that many MSK disorders are caused by myofascial pain, a summary of NT effects and mechanisms in modern scientific terminology can provide practitioners with clearer clinical reasoning to justify their treatment using NT to treat myofascial pain and MSK disorders. This review summarised 4 main effects according to different anatomic levels of the human body attributed to NT. They are myofascial effect, supraspinal effect, spinal effect and local effect. The mechanisms responsible for each of these effects, respectively, are suppressing motor end plate activity, descending inhibition mechanism, control gate mechanism and axon reflex.

Keywords Needling therapy, Acupuncture, Dry needling, Myofascial pain, Musculoskeletal disorders, Mechanisms, Effects

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Graphical Abstract

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Background

What is needling therapy (NT)

- NT is an umbrella term for any modality involving the use of needles inserting into the soft tissue to treat musculoskeletal (MSK) problems.
- Two common forms of NT: acupuncture and dry needling (DN)

Rationales

- Myofascial pain is a major contributor to many MSK disorders.
- NT is growing popularity to treat MSK disorders.
- Systematic reviews and meta-analyses have proven effectiveness of NT to treat MSK disorders.
- Acupuncture vs. DN
"Acupuncture" is hard to be validated by modern science as involving Chi, "Yin & Yang" and "meridians" - all the ambiguous terms, and there is no unified mechanisms for DN documented well.
- There is a need to explore the mechanisms behind NT including acupuncture explained in modern scientific terminology to justify the use of it.

Objectives

- To summarise and categorise NT effects according to different anatomic levels of the human body and link each effect to the specific underlying mechanism.

Methods

Search sequence "T1 AND T2 AND T3" was searched through electronic databases to identify potential studies

Where, T1 = conditions, T2 = interventions, T3 = effects/mechanisms
CBEM appraisal tool were used for appraising RCTs, Mirror for NRCTs and SARN for narrative reviews

Results (No. of studies)

- N = 27 studies identified of which
- 1 RCT
 - 4 NRCTs
 - 11 narrative reviews
 - 6 theoretical articles
 - 1 observational study
 - 3 experimental studies (neither NRCT nor RCT)
 - 1 expert opinion

Results (Effects)

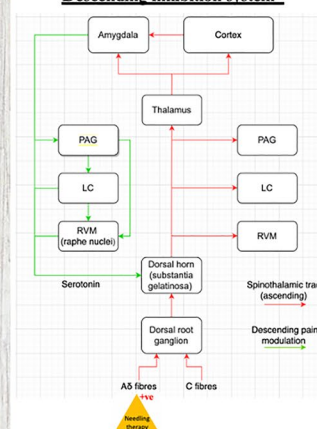
NT effects were summarised into 4 anatomic levels of the human body

Effects	Underlying Mechanisms
• Myofascial effect	• Suppressing motor end plate activity
• Supraspinal effect	• Descending inhibition system*
• Spinal effect	• Control gate mechanism - presynaptic inhibition*
• Local effect	• Axon reflex

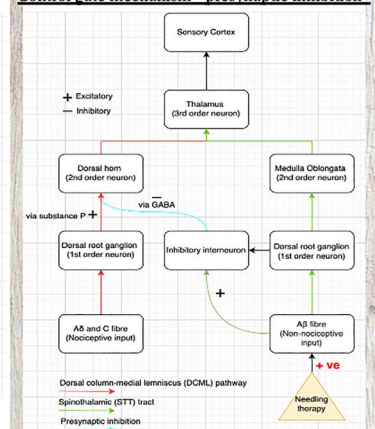
Implication

- By summarising and categorising the four effects and mechanisms into different anatomic levels of the human body, it can provide practitioners with clearer clinical reasoning to justify their treatment using NT to treat myofascial pain and MSK disorders.

Descending inhibition system*



Control gate mechanism - presynaptic inhibition*



1 Introduction

Musculoskeletal (MSK) disorders are defined as conditions affecting muscles, ligaments, tendons, bones and nerves acutely and chronically [1]. Data from the Global Burden of Disease shows that MSK disorders are the most prevalent category of health issue affecting nearly 1.71 billion people worldwide [2]. In the United Kingdom, almost one third of the population suffer from MSK disorders [3]. The World Health Organization has defined MSK disorders as the leading contributor to disability worldwide [4]. While the most prevalent MSK disorders is low back pain [4], 85% of low back pain is caused by myofascial pain [5], and myofascial pain accounts for 30–85% of other different MSK disorders [6]. A systematic review conducted by Babatunde et al. [7] has shown that exercise therapy, psychosocial interventions and pharmacological treatments are effective to treat MSK disorders. However, these commonly

practised modalities appear to be insufficient to lower the high prevalence of MSK disorders.

Needling therapy (NT), therefore, starts to gain its popularity as a new modality to treat MSK disorders due to growing evidence to prove its efficacy. NT is an umbrella term for any modality involving the use of needles inserting into the soft tissue to treat MSK pain and conditions [8]. There are two common forms of NT: acupuncture and dry needling (DT). While DT is defined by the American Physical Therapy Association (APTA) as a skilled intervention involving the use of a filiform needle to penetrate the skin to stimulate underlying myofascial points, muscular and connective tissues to treat MSK pain and movement impairments [9], acupuncture (translated as needle penetration) is defined in TCM as the use of small needles inserted into the specific points of the body (acupoints points) to treat bodily problems [10]. Two systematic reviews have shown effectiveness

of NT to treat low back pain (LBP) [11, 12]. One systematic review and meta-analysis has shown effectiveness of NT combined with stretching to treat myofascial trigger points (TrP) [13]. Other systematic reviews and meta-analyses have also shown effectiveness of NT to treat temporomandibular disorder [14], frozen shoulder [15], stiff neck [16] and plantar fasciitis [17]. These modern evidences proven in the West demonstrate the effectiveness of NT to treat MSK disorders, and the first form of NT—acupuncture—actually has been practised in the East for almost 3000 years [18].

The first documentation of acupuncture was in “Nuangdi Neijing”, which is translated as “the Inner Canon of the Yellow Emperor”. It was a book written in about 2000 years ago in the Han Dynasty and is the fundamental doctrine for traditional Chinese Medicine (TCM) [19]. This book has documented the mechanism of acupuncture to explain how it works to treat all bodily pains and conditions. The underlying mechanism of acupuncture, according to the book, is to balance the “Yin” and “Yang” of a human body to restore internal harmony. “Yin” and “Yang” are both important opposing concepts in TCM. While “Yin” symbolises female, cold, quiet, etc., “Yang” symbolises male, warm, movement, etc. The human body has 20 meridian channels, where contain “Chi” to flow. “Chi”, another important concept in TCM, can be translated as “vital energy”. While “Chi” flows through the meridian channels, and “blood” flows through the blood vessels. Blood contributes to “Yin”, and “Chi” contributes to “Yang”. Acupuncture has an effect on “Yang”. Since “Yin” and “Yang” has to be balanced well in the human body to achieve internal harmony, all bodily pains and conditions occur once human body loses internal harmony (i.e., “Yin” and “Yang” are not balanced), and acupuncture works by modulating “Yang” in the human body to achieve internal harmony. Once internal harmony is achieved, all diseases are gone (Fig. 1).

However, the mechanism of acupuncture explained by TCM all depends on “Chi”. Without the proof of the existence of “Chi”, the whole theory does not stand. Unfortunately, 2000 years later, “Chi” still has not been proven yet. This theory, therefore, is not likely to be accepted in the current literature. As a result, the mechanisms behind NT including acupuncture explained in modern scientific terminology need to be explored to justify the use of it. In addition, NT has grown in popularity over the last 20 years to treat myofascial pain and MSK disorders [20]; however, no unified theories and mechanisms have well been reviewed and documented in the current literature—most of them are rather equivocal. This review is the first study aiming to summarise and categorise NT effects according to different anatomic levels of the human body and to link each effect to the specific underlying mechanism to treat myofascial pain

State of internal harmony = State of disease free

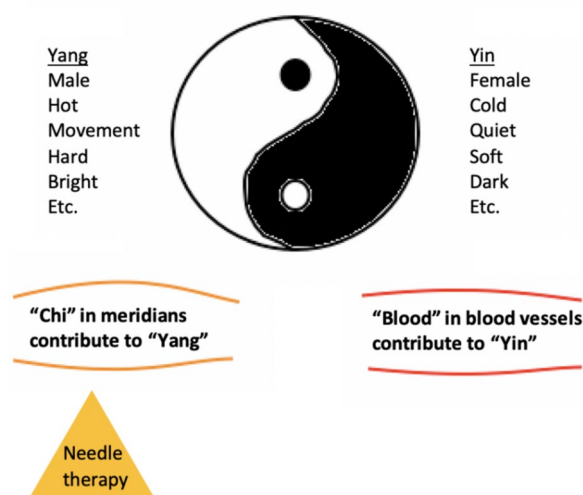


Fig. 1 State of internal harmony with concept of Yin and Yang according to TCM

and MSK disorders. The terms “effect” and “mechanism” are causally related, so by unifying them, it can provide practitioners with clearer clinical reasoning to justify their treatment using NT to treat myofascial pain and MSK disorders.

2 Methods

The methodology of narrative review was adopted for the conduction of this review as the scope of this study is of a border topic with an intention of synthesising and summarising the current literature on effects and mechanisms of NT (acupuncture and dry needling) for treating wider population of conditions which are myofascial pain and MSK disorders.

Four electronic databases (AMED, CLINAL, Cochrane Library and Medline) were searched to identify the potential studies. External potential studies were also identified through manual searching of citation from the included studies. Three main themes comprising the search strategy were defined (conditions, interventions and effects/mechanisms). Synonyms and words of similar concepts in each theme were connected using Boolean operator “OR”. The final search sequence was then created by connecting each theme with Boolean operators (“AND”). Detailed search strategy was listed in appendix 1. Exclusion criteria were duplicate studies, studies without full text, not in English and conducted before 2005 and studies with non-available references. In addition, this review is only interested in NT that does not involve injection of substance; therefore, any studies focus on wet needling were also excluded. The search years were from 2005 to 2025 (manual citation search not limited to years). Studies were not limited to RCTs, systematic

reviews and meta-analyses, but the source types were limited to academic journals only. The studies were only selected if they have reported and mentioned any effects or mechanisms of NT for treating myofascial pain or MSK disorders.

There were 27 studies being identified in total (Fig. 2), including 1 randomized control trial (RCT) [21], 4 non-randomized control trials (NRCTs) [22–25], 11 narrative reviews [26–36], 6 theoretical articles [37–42], 1 observational study [43], 3 other experimental studies [44–46] and 1 expert opinion [47]. Quality appraisal was completed for the identified RCTs, NRCTs and narrative reviews. Centre for Evidence-Based Medicine (CEBM)

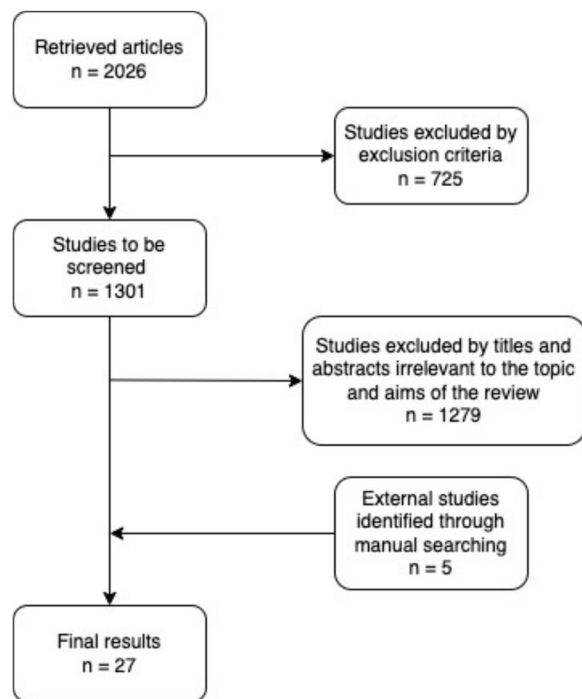


Fig. 2 Study screen flow diagram

RCT critical appraisal sheet was applied to appraise the identified RCT studies [48]; methodological Items for Non-Randomized Studies (MINORS) instrument was applied to appraise the identified NRCT studies [49]; the Scale for the Assessment of Narrative Review Articles (SARN) was applied to appraise the identified narrative reviews [50]. The appraisal results are listed in Appendix 2–4.

3 Four Main Needling Therapy Effects and Underlying Mechanisms

3.1 Myofascial Effect and Suppression of Motor End Plate Activity

One of the major effects attributed to NT in the treatment of MSK pain is myofascial effect. This effect, as implied by its name, occurs at the myofascial level of the human body. To understand how NT produces this effect, it is essential to first introduce and understand the pathophysiology of TrP (Fig. 3).

The pathophysiology of TrP can be explained by the integrated trigger point hypothesis (ITPH) [26]. It suggests that the dysfunction of motor endplate is the cause of TrP [37]. The dysfunction of motor endplate leads to

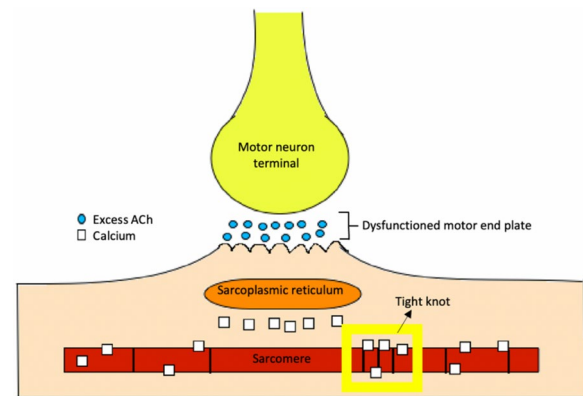


Fig. 4 Dysfunctional motor end plate activity

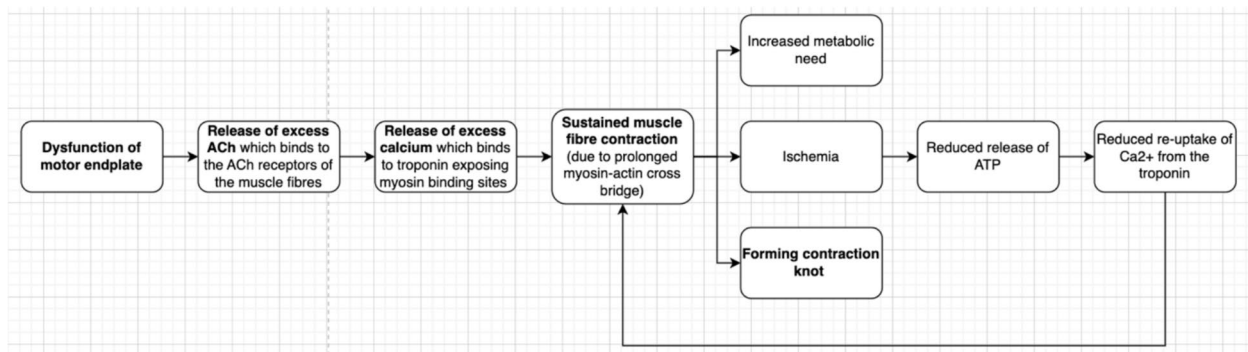


Fig. 3 Pathophysiology of TrP

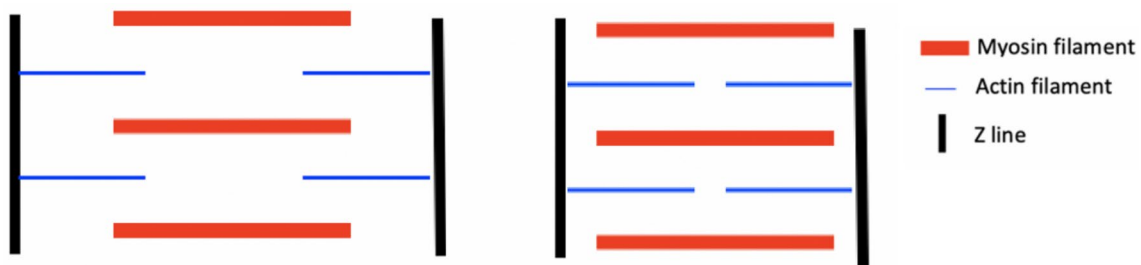


Fig. 5 Muscle sliding filament theory—actin and myosin filament

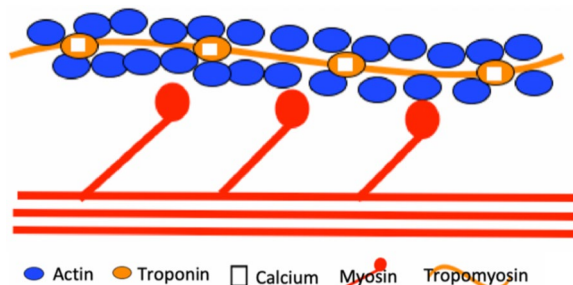


Fig. 6 Muscle sliding filament theory—actin myosin cross bridge

the excess release of acetylcholine (ACh), which is the culprit of forming TrP [38] (Fig. 4).

ACh is the neurotransmitter travelling along the motor end plate, binding to the ACh receptors of the muscle fibres. Once the binding occurred, calcium is released from the sarcoplasmic reticulum (SR) and attaches onto troponin to expose the myosin binding sites of actin so that myosin head can bind to actin to form the cross bridge. The sliding between the actin–myosin filaments can then bring two adjacent Z lines close together, hence shortening the sarcomere. This sliding mechanism is known as muscle sliding filament theory [27, 51] (Figs. 5, 6). However, whenever there is excessive release of ACh, excess calcium is then

released from SR, causing the myosin binding sites constantly to expose themselves to form the constant crossbridge. Since actin and myosin filaments slide one pass each other, sarcomere is then shortened sustainably, causing prolonged muscle fibre contraction. Eventually, contraction knots (i.e., TrP) are formed [43] (Fig. 4).

Sustained muscle fibre contraction not only enhances metabolic demand, but also may cause ischemia [28, 29]. Since oxygen is required to create adenosine triphosphate (ATP), ischemia depletes ATP. ATP is needed to release myosin head from binding to actin to break the crossbridge according to the muscle sliding filament theory [51], so a lack of ATP means crossbridge forms constantly. As long as there is crossbridge formation, the reuptake of Ca^{2+} back to the SR is paused as Ca^{2+} is required to form the crossbridge; therefore, the constant crossbridge formation leads to shortened sarcomere and sustained muscle fibre contraction, eventually forming TrP [43].

NT works by breaking this envious cycle (Fig. 7). This is known as myofascial effect. There are three sub mechanisms to explain how myofascial effect works. First, NT may suppress the motor end plate activity (i.e., depleting ACh level) [21, 30]. It is found that this suppression effect is more pronounced when rapid dry needling is applied to elicit multiple lateral twitch responses (LTRs), which

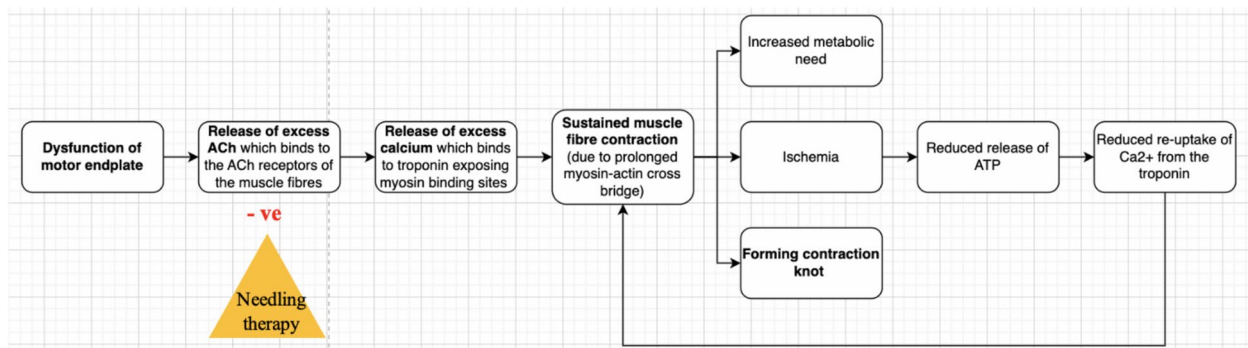


Fig. 7 Breaking the envious cycle—reducing release of ACh at the motor endplate

The second effect attributed to NT in the treatment of MSK pain is supraspinal effect (Fig. 8). As its name suggested, this effect happens at the supraspinal level of the

Nociceptive input travels along the spinothalamic tract ascending into the sensory cortex to perceive pain. There are two sensory fibres forming the spinothalamic tract picking up nociceptive signals: 1) A-delta ($A\delta$) fibres and C fibres. While $A\delta$ fibres are high threshold afferent fibres; C fibres are of low threshold [52]. Therefore, any low intensity nociceptive inputs will stimulate C fibres. Since input from NS is of high intensity comparatively, it will stimulate $A\delta$ fibres. In terms of myelination and conduction velocity, $A\delta$ fibres are moderately myelinated with conduction velocity between 1 and 4 m/s; C fibres are non-myelinated with maximum conduction velocity to be less than 1 m/s [24]. Therefore, any input travelling



along the A δ fibres will be more rapid than that travelling along the C fibres. Descending inhibition occurs once A δ fibres are activated [32]; therefore, if both A δ fibres and C fibres are stimulated at the same time, it makes sense that descending pain modulation should occur quicker than the pain ascending to the sensory cortex.

Once the descending inhibition is activated by NT, a chain of neurons which descending the length of the spinal cord are activated as well [53]. There are two key structures of descending inhibitory system: periaqueductal gray (PAG) and amygdala [39, 54]. While amygdala is part of limbic system, playing important roles in emotional responses, stress and anxiety, PAG modulates rostral ventromedial medulla (RVM) to release serotonin into an area called substantia gelatinosa of the dorsal horn [54]. Once the serotonin hits to the dorsal horn, it will bind to the interneurons. The interneurons then secrete opioids (enkephalin/dynorphin) to the terminals of the 1st order neuron (C fibres), where opioid receptors are present. These endogenous opioids inhibit the release of substance P (SP) from the 1st order neuron to the 2nd order neuron. Since SP is the neurotransmitter carrying nociceptive signal along the spinothalamic tract, the inhibition of release of it means there is less nociceptive signal ascending along the tract to the sensory cortex, and as a result, less pain being perceived [32]. To sum up the supraspinal effect produced by NT, the main principle is to “use pain to curb pain”.

3.3 Spinal Effect and Control Gate Mechanism

The third effect attributed to NT in the treatment of MSK pain is spinal effect which can be explained by gate control mechanism [40]. The key concept of gate control mechanism is presynaptic inhibition through which the “pain gate” is closed, hence causing analgesic effect (Fig. 9).

There are two main tracts involved in gate control mechanism: spinothalamic tract (STT) and dorsal column-medial lemniscus (DCML) pathway. STT transmits nociceptive signals from either A δ fibres or C fibres all the way to the sensory cortex, while DCML transmits non-nociceptive signals from A-beta (A β) fibres all the way to the sensory cortex [55]. In the spinothalamic tract, either A δ or C fibre picks up the nociceptive input, and SP acts as neurotransmitter to carry this nociceptive signal from the 1st order neuron to the 2nd order neuron in the dorsal horn. Once the nociceptive input reaches cortex, pain is perceived. The synapse between the 1st and 2nd order neurons is where the pain gate located. As long as this gate is closed, SP will be limited to travel through. Although it is impossible to fully close this gate, it conveys an idea that how the interruption of the transmission of SP within the synapse can lower pain perception

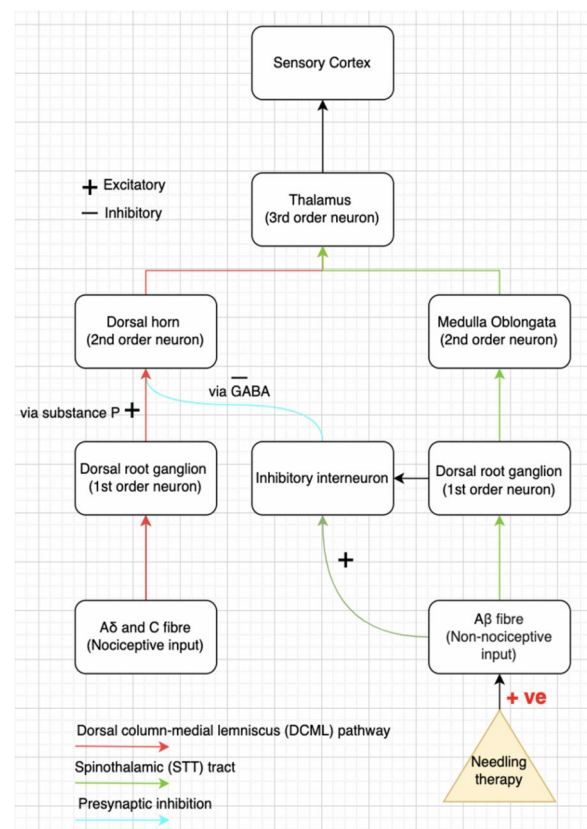


Fig. 9 Control gate mechanism—presynaptic inhibition

created by the cortex. One way to achieve this is presynaptic inhibition.

NT can activate presynaptic inhibition to close the pain gate (i.e., interrupting the nociceptive signal travelling between the 1st and 2nd order neurons) by stimulating the A β fibre [33]. Upon the stimulation, the A β fibre stimulates the inhibitory interneuron, the axon of which then secretes gamma-aminobutyric acid (GABA) as neurotransmitter to the axon of the 1st order neuron of the STT. GABA is inhibitory neurotransmitter, which can inhibit the release of SP released from the 1st order neuron to the 2nd of STT [34], hence less nociceptive signals travelling to the brain to create pain perception.

3.4 Local Effect and Axon Reflex

Local effect occurs whenever a needle is inserted into the local injured muscle tissue (Fig. 10). Axon reflex is the underlying mechanism responsible for the local effect induced by NT and is a primary physiological response to peripheral injury [41].

During axon reflex the nerve sensory fibres are stimulated to release neuropeptides to cause vasodilation and increased local circulation [41, 42]. Insertion of needles can induce a controlled microtrauma (i.e., peripheral

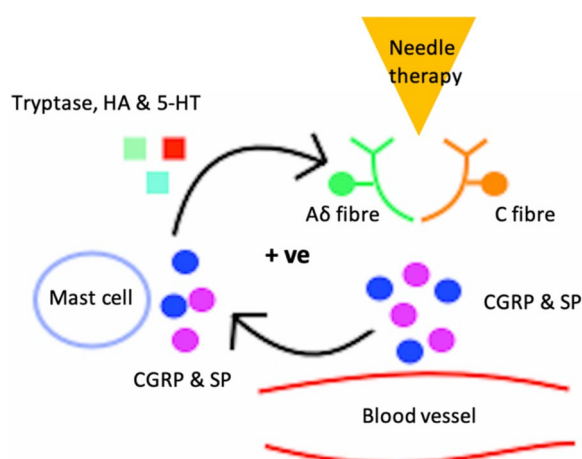


Fig. 10 Local effects

injury) that causes axon reflex [35, 36]. When a needle inserts into the tissue, the Aδ or C fibres are activated to release vasoactive neuropeptides, primarily CGRP and SP [42]. Both CGRP and SP are abundantly expressed in the nerve fibres and are potent vasodilators to cause increasing blood flow superficially and deeply to the local muscle tissue [25, 45, 46]. Therefore, after an individual receives needling treatment, the tissue often responds with flushing bright red which implies the increasing blood flow in the local area caused by axon reflex [41]. The increasing blood flow of the injured muscle tissue serves as a key factor to promote healing, hence causing pain reduction.

In addition, the CGRP and SP released from the nerve fibres during axon reflex induced by insertion of needles not only act on the blood vessels in the local tissue but also act on the mast cells in the local tissue to release 3 neurotransmitters: tryptase, histamine (HA) and serotonin (5-HT) [25]. These 3 neurotransmitters enhance the axon reflex induced by insertion of needles to intensify the local effect (i.e., vasodilation) [25].

4 Limitations

One limitation of this narrative review is that it was conducted by single screening approach for study selection due to limited resource, so it might be prone to bias; however, it requires fewer resources to conduct, offering potential shortcuts for rapid reviews [56]. To minimise the effect due to single screening approach, different quality appraisal tools were applied to assess the quality of studies of different types. However, only three types of study (RCTs, NRCTs and narrative reviews) were appraised as there are no appraisal tools available for other types of studies. All the judging criteria, scoring and results are tabulated in Appendix 2, 3, 4.

Although the study selection was not limited to RCTs, systematic review and meta-analyses which are of high standard for effective research [57], there were no systematic review and meta-analysis identified in this review, and only 1 RCT study was identified; the rest types of studies are of the lower hierarchy of evidence [57]. In addition, 1 identified NRCT [24] and 2 other experimental studies [35 & 54] were animal trials (rats); however, laboratory rats are ideal animal models for biomedical research as about 95% of their genes are shared with humans [58]. Furthermore, this review focuses only on two common forms of NT (acupuncture and DN), while wet needling which involves the addition of an injection of analgesic substance [59] is not explored in this review.

One final limitation of this review is that this review explores and documents the effects and mechanisms of NT (including acupuncture) in western scientific terminology. The results of this review might not be compatible with or transferable to the TCM literature, which, however, might cause controversy or conflicting ideology for traditional TCM practitioners who perform the modality. However, this review does not intend to reach consensus, change the theory ingrained in TCM nor change the ideology of those TCM practitioners who perform the modality but opens an alternative way of thinking for the practitioners to justify their selection of the modality.

5 Conclusion

This review summarised and categorised four main effects and the underlying mechanisms of NT according to different anatomic levels of the human body. They are myofascial effect, supraspinal effect, spinal effect and local effect. Each effect occurs at the specific anatomic level of the human body as the name implies. For myofascial effect, NT suppresses the release of ACh at the motor end plate within the myofascial tissue, reducing motor end plate activity. For supraspinal effect, NT activates the descending inhibition system at the supraspinal level, promoting the release of endogenous opioids that travel down along the spinal cord to alleviate pain (i.e., use pain to curb pain). For spinal effect, NT triggers presynaptic inhibition at the spinal level, closing the “pain gate” to reduce nociceptive signal transmission. Finally, the mechanism behind local effect is that NT induces axon reflex as a response of controlled microtrauma at the local injured tissue, creating vasodilation to accelerate tissue healing. By summarising and categorising these four effects and mechanisms according to different anatomic levels of the human body, it can provide practitioners with clearer clinical reasoning to justify their treatment using NT to treat myofascial pain and MSK disorders.

Appendix 1

See Table 1.

Table 1 Search strategy

Search strategy	
Theme 1—conditions	"Myofascial pain" OR "Trigger point*" OR "Musculoskeletal pain" OR "Musculoskeletal disorder*" OR "Musculoskeletal condition*" OR "Musculoskeletal disease"
Theme 2—interventions	"Needling therapy" OR "Dry needling" OR Acupuncture
Theme 3—effects/mechanisms	Effect* OR Mechanism*
Final search sequence	"Theme 1" AND "Theme 2" AND "Theme 3"

Appendix 2

See Table 2.

Table 2 Quality appraisal result (CBEM for RCTs)

CEBM (RCTs)						
Studies	Randomised allocation	Group similarity	Group treated equally	All subjects accounted for in groups	Blinding	Rating
Abbaszadeh-Amirdehi et al. (33)	Yes	Yes	Yes	Yes	No	4 (80%)

Appendix 3

See Table 3.

Table 3 Quality appraisal result (MINOR for NRCTs)

MINOR (NRCTs)													
Studies	A clearly stated aim	Inclusion of consecutive patients	Prospective collection of data	Endpoints appropriate to the aim of the study	Unbiased assessment of the study endpoint	Follow-up period appropriate to the aim of the study	Loss to follow up less than 5%	Prospective calculation of the study size	Adequate control group	Contemporary groups	Baseline equivalence of groups	Adequate statistical analyses	Total scores
Graeff et al. (38)	2	2	0	2	0	1	0	0	2	0	2	2	13 / 24 (54%)
Chou et al. (39)	2	2	0	2	2	0	0	0	2	0	2	2	14 / 24 (58%)
Chen et al. (41)	2	0	1	2	0	0	0	0	NA	NA	NA	NA	5 / 16 (31%)
Wu et al. (56)	2	0	1	2	0	0	0	0	1	0	1	2	9 / 24 (38%)

Appendix 4

See Table 4.

Table 4 Quality appraisal result (SANRA for Narrative reviews)

SANRA (Narrative reviews)							
Studies	Justification of the importance	Aims of the review	Literature search	Referencing	Scientific reasoning	Presentation of data	Sumscore
Bron & Dommerholt (24)	1	0	0	2	1	2	6 / 12 (50%)
Squire (27)	0	1	0	2	2	2	7 / 12 (58%)
Dommerholt et al. (30)	2	2	0	2	2	2	10 / 12 (83%)
Shah (31)	1	1	0	2	2	2	8 / 12 (67%)
Perreault et al. (34)	2	2	2	2	2	2	12 / 12 (100%)
Lv et al. (36)	2	2	0	2	2	1	9 / 12 (75%)
Kwon et al. (42)	2	1	0	2	2	2	9 / 12 (75%)
Cagnie et al. (48)	2	2	0	2	1	2	9 / 12 (75%)
Guo & Hu (49)	1	2	1	2	2	2	10 / 12 (83%)
Cheng (52)	0	1	1	2	1	2	7 / 12 (58%)
Jin et al. (53)	1	1	0	1	2	1	6 / 12 (50%)

Abbreviations

ACh	Acetylcholine
ATP	Adenosine triphosphate
CEBM	Centre for evidence-based medicine
CGRP	Calcitonin gene-related peptide
DCML	Dorsal column-medial lemniscus
DT	Dry needling
GABA	Gamma-aminobutyric acid
HA	Histamine
LBP	Low back pain
LTR	Lateral twitch responses
MINORS	Methodological items for non-randomized studies
MSK	Musculoskeletal
NT	Needling therapy
PAG	Periaqueductal gray
RVM	Rostral ventromedial medulla
SARN	Scale for the assessment of narrative review articles
SP	Substance P
SR	Sarcoplasmic reticulum
STT	Spinothalamic tract
TCM	Traditional Chinese Medicine
TrP	Trigger point
5-HT	Serotonin

Author Contributions

CMT confirms sole responsibility for the following: research idea, study selection, data analysis and synthesis, graphics creation, interpretation of results and manuscript writing.

Funding

None.

Availability of Data and Materials

No datasets were generated or analysed during the current study.

Declarations**Conflict of Interest**

The authors declare no competing interests.

Ethical Approval and Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

Clinical Trial Number

Not applicable.

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